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Fujimura et al.

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(54) **INFORMATION PROCESSING DEVICE, INFORMATION PROCESSING METHOD, AND RECORDING MEDIUM**

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G06F 18/21 (2023.01)
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CPC ***G06N 3/08*** (2013.01); ***G06F 18/21*** (2023.01); ***G06F 18/2415*** (2023.01);
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CPC G06N 3/08; G06N 20/00; G06N 5/04
See application file for complete search history.

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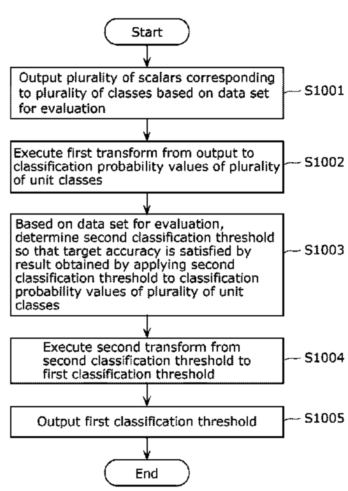
(74) *Attorney, Agent, or Firm* — Wenderoth, Lind & Ponack, L.L.P.

(57) **ABSTRACT**

An information processing device includes a processor. The processor obtains a first classification threshold for classifying data into at least one of a plurality of classes, and outputs a classification result of classifying the data into at least one of the plurality of classes based on an output of a trained classification model and the first classification threshold. The first classification threshold is obtained by a second transform performed on a second classification threshold, the second transform being an inverse transform of a first transform. The first transform corresponds to transforming the output of the trained classification model into a classification probability value of each of a plurality of unit classes constituting the plurality of classes. The second classification threshold is set based on the classification probability values of the plurality of unit classes.

6 Claims, 10 Drawing Sheets



Related U.S. Application Data

(60) Provisional application No. 62/787,576, filed on Jan. 2, 2019.

(51) **Int. Cl.**

G06F 18/2415	(2023.01)
G06F 18/2431	(2023.01)
G06N 5/04	(2023.01)
G06N 20/00	(2019.01)
G06V 10/764	(2022.01)

(52) **U.S. Cl.**

CPC ***G06F 18/2431*** (2023.01); ***G06N 5/04*** (2013.01); ***G06N 20/00*** (2019.01); ***G06V 10/764*** (2022.01)

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FIG. 1

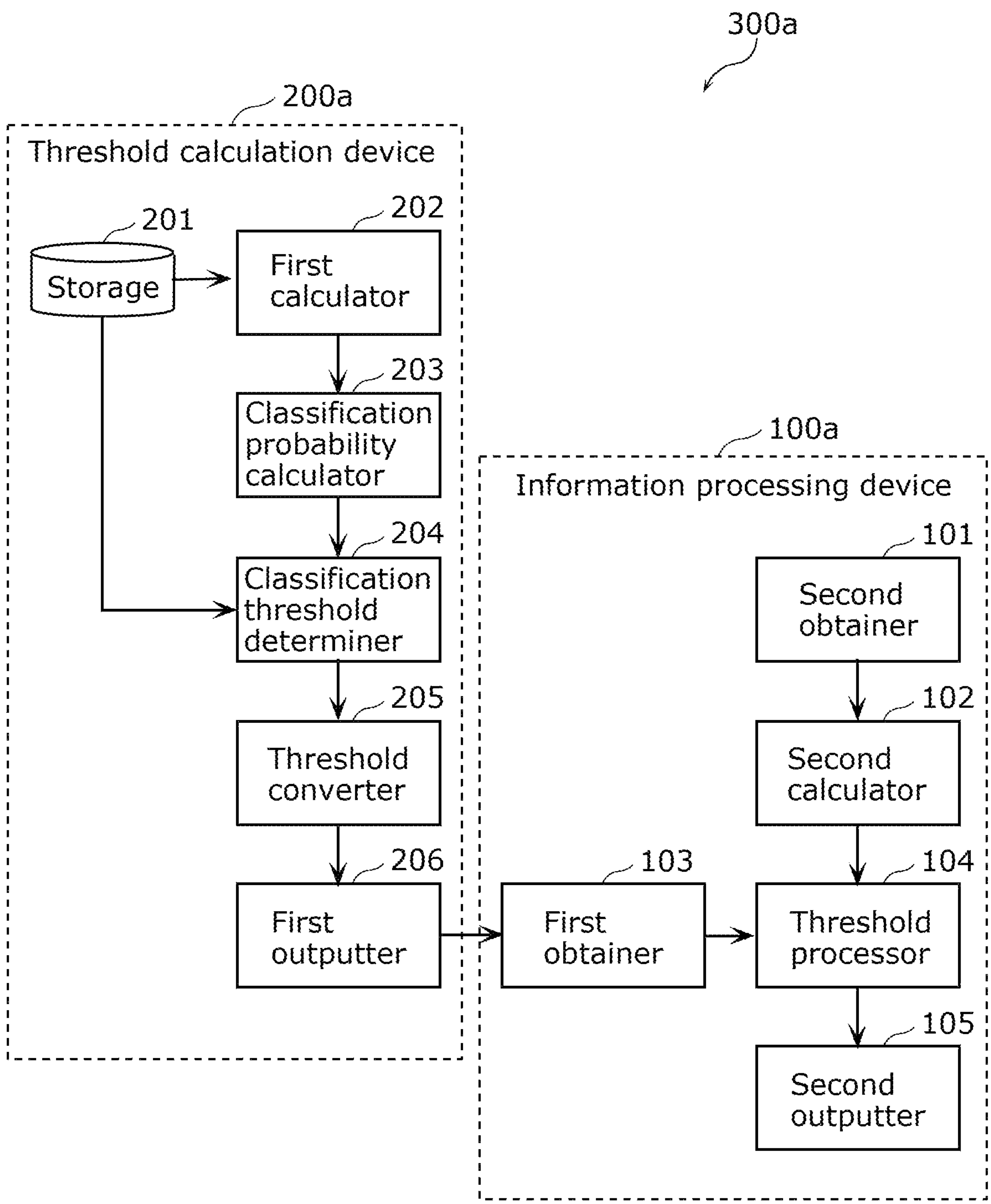

300a
200a
Threshold calculation device
201
Storage
202
First calculator
203
Classification probability calculator
204
Classification threshold determiner
205
Threshold converter
206
First outputter
100a
Information processing device
101
Second obtainer
102
Second calculator
103
First obtainer
104
Threshold processor
105
Second outputter

FIG. 2

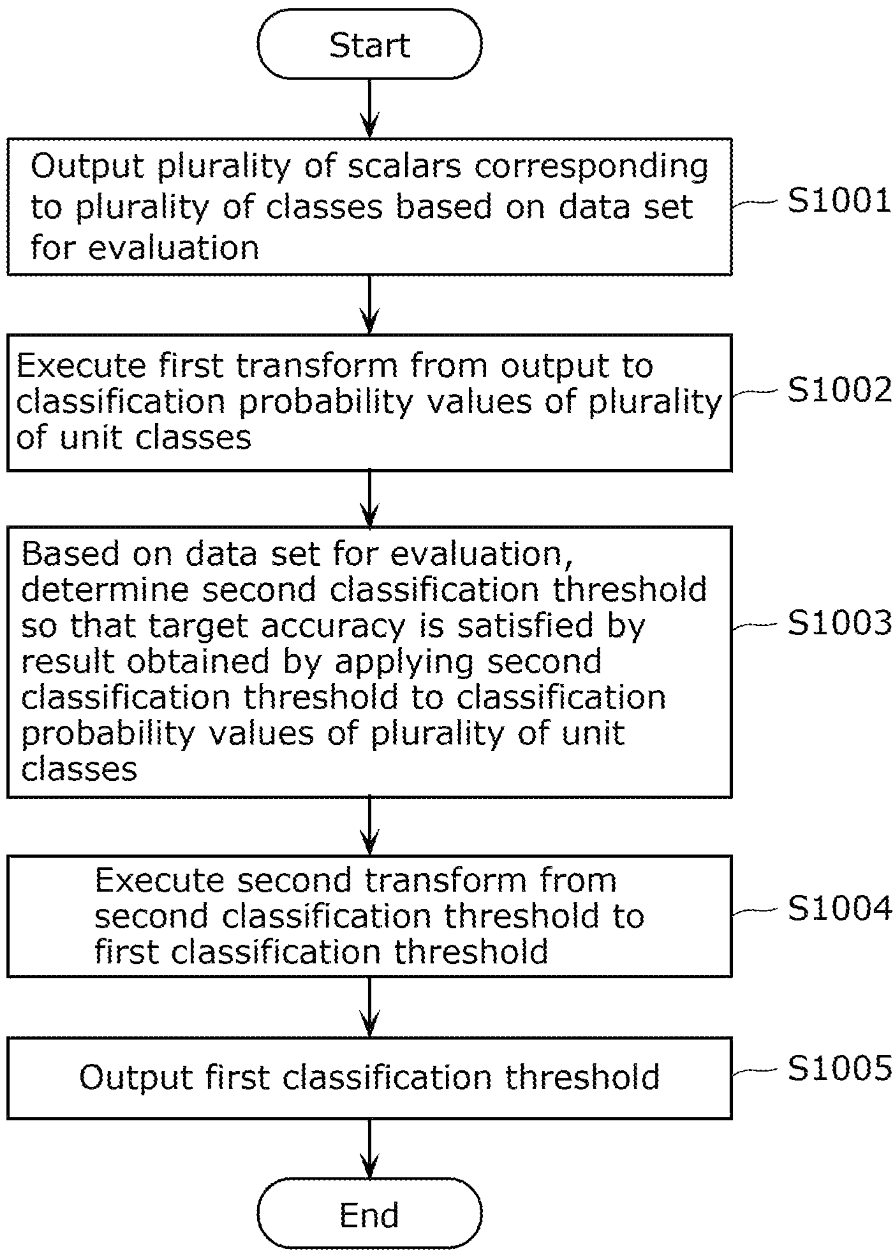

Start
Output plurality of scalars corresponding to plurality of classes based on data set for evaluation
S1001
Execute first transform from output to classification probability values of plurality of unit classes
S1002
Based on data set for evaluation, determine second classification threshold so that target accuracy is satisfied by result obtained by applying second classification threshold to classification probability values of plurality of unit classes
S1003
Execute second transform from second classification threshold to first classification threshold
S1004
Output first classification threshold
S1005
End

FIG. 3

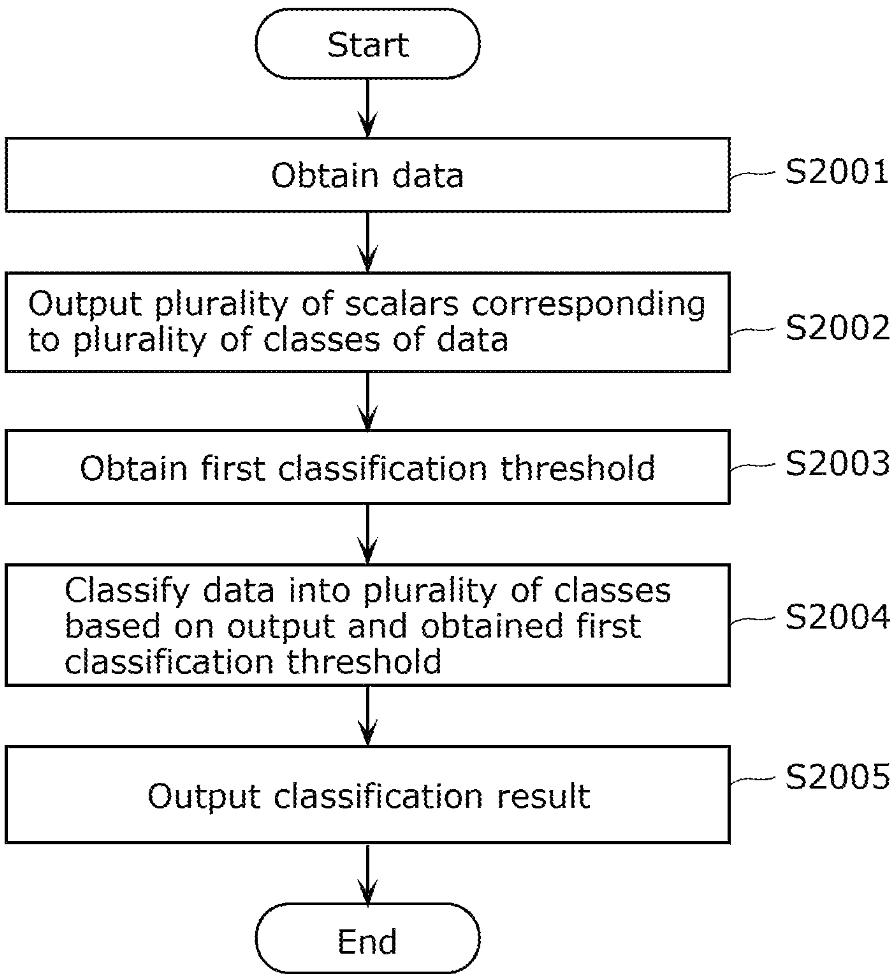

Start
Obtain data
S2001
Output plurality of scalars corresponding to plurality of classes of data
S2002
Obtain first classification threshold
S2003
Classify data into plurality of classes based on output and obtained first classification threshold
S2004
Output classification result
S2005
End

FIG. 4

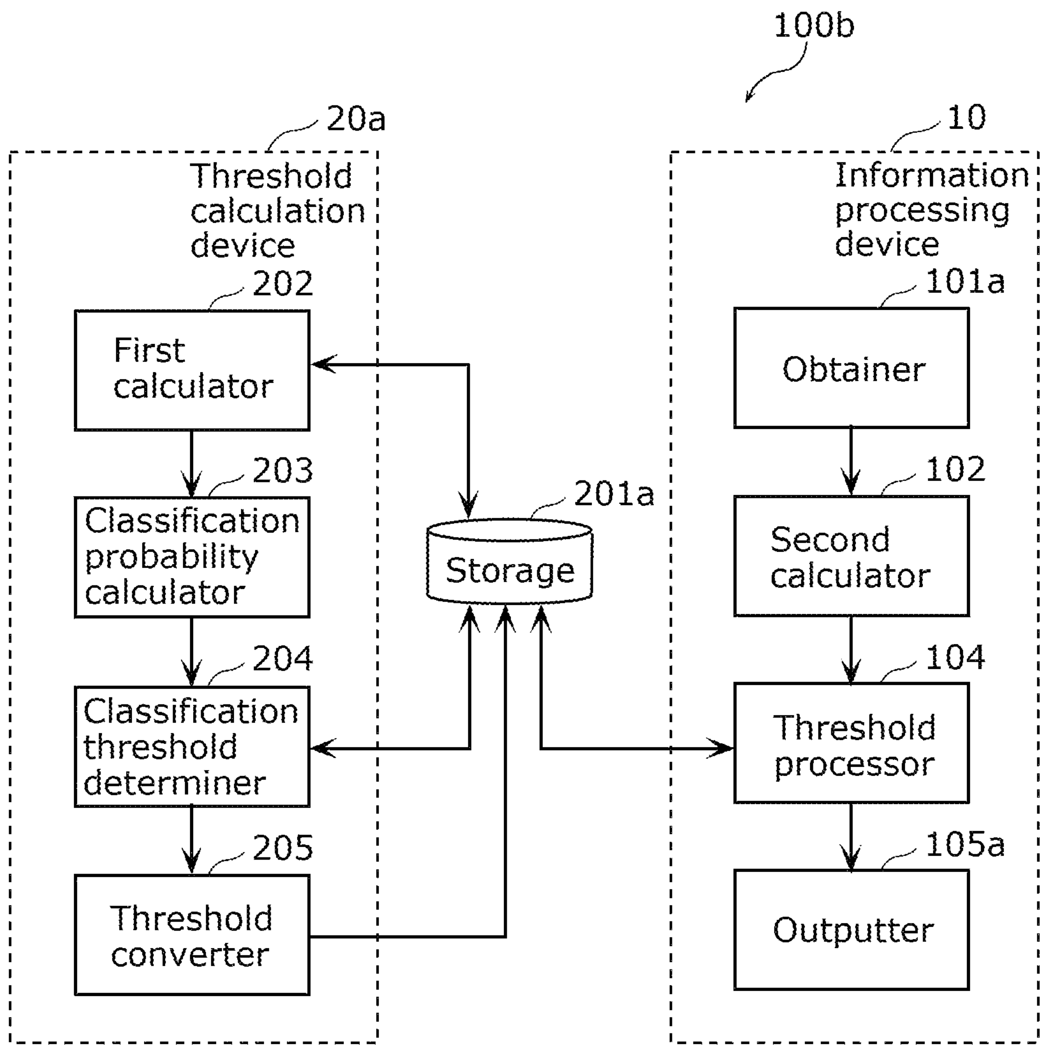

100b
20a
Threshold calculation device
202
First calculator
203
Classification probability calculator
204
Classification threshold determiner
205
Threshold converter
201a
Storage
10
Information processing device
101a
Obtainer
102
Second calculator
104
Threshold processor
105a
Outputter

FIG. 5

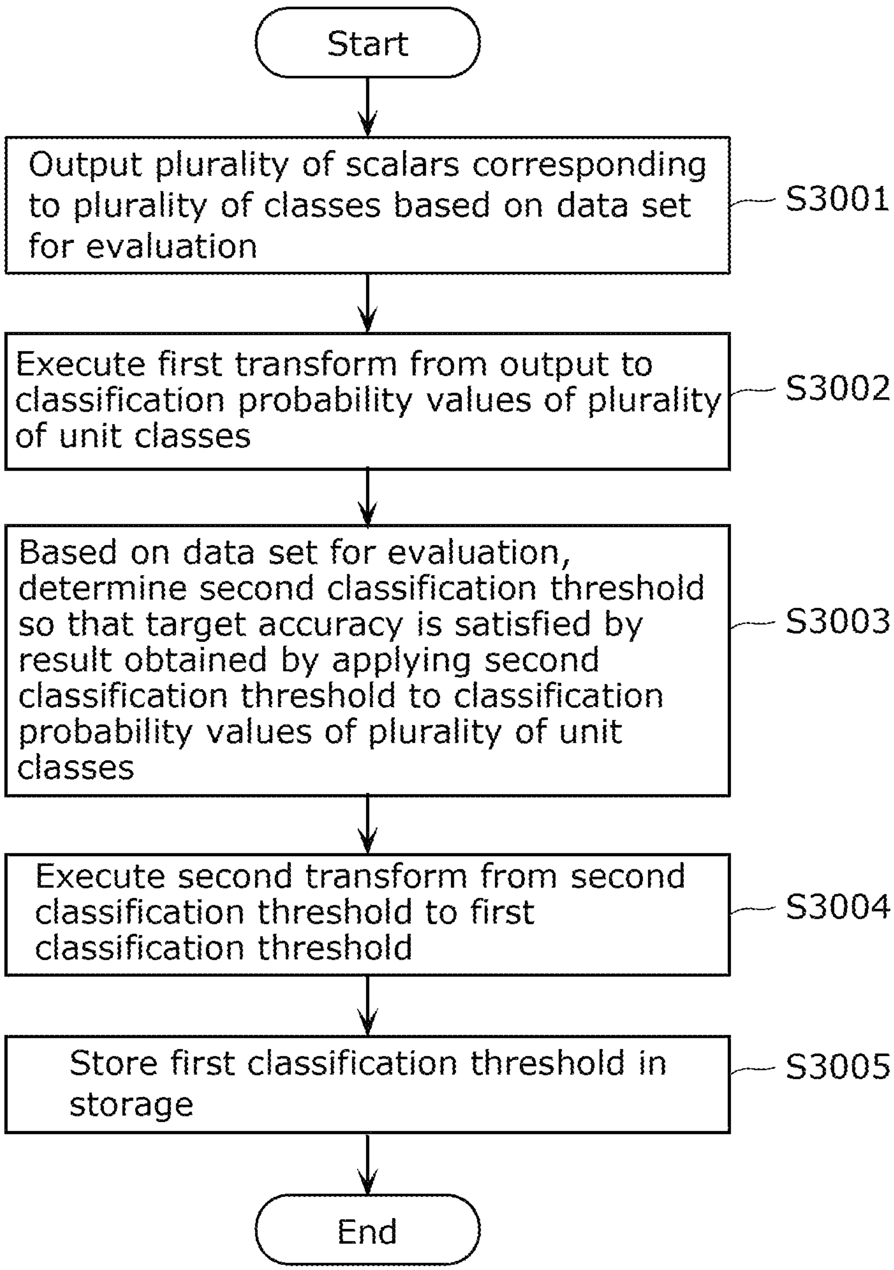

Start
Output plurality of scalars corresponding to plurality of classes based on data set for evaluation
S3001
Execute first transform from output to classification probability values of plurality of unit classes
S3002
Based on data set for evaluation, determine second classification threshold so that target accuracy is satisfied by result obtained by applying second classification threshold to classification probability values of plurality of unit classes
S3003
Execute second transform from second classification threshold to first classification threshold
S3004
Store first classification threshold in storage
S3005
End

FIG. 6

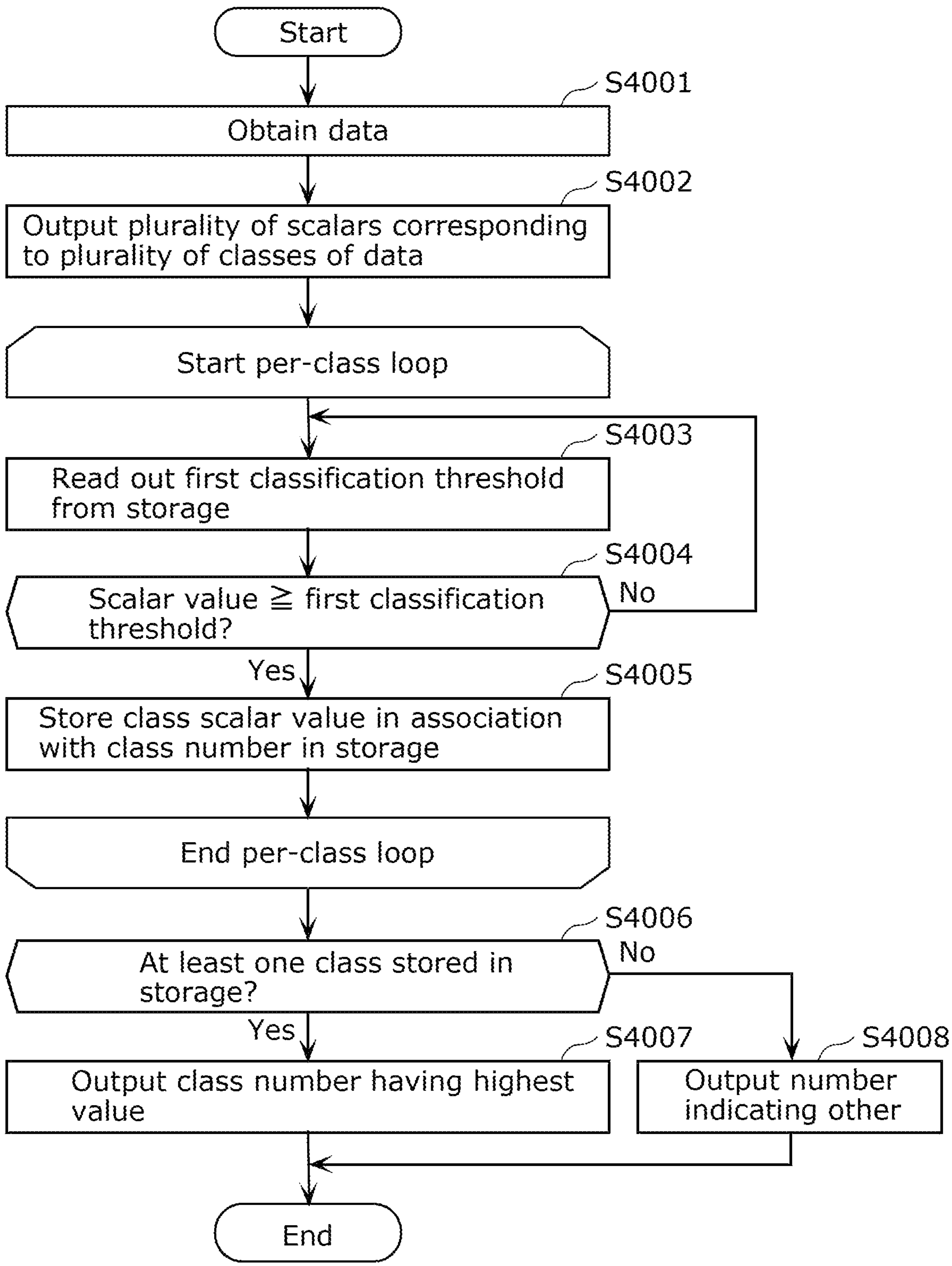

Start
S4001
Obtain data
S4002
Output plurality of scalars corresponding to plurality of classes of data
Start per-class loop
S4003
Read out first classification threshold from storage
S4004
Scalar value ≧ first classification threshold?
No
Yes
S4005
Store class scalar value in association with class number in storage
End per-class loop
S4006
At least one class stored in storage?
No
Yes
S4007
Output class number having highest value
S4008
Output number indicating other
End

FIG. 7

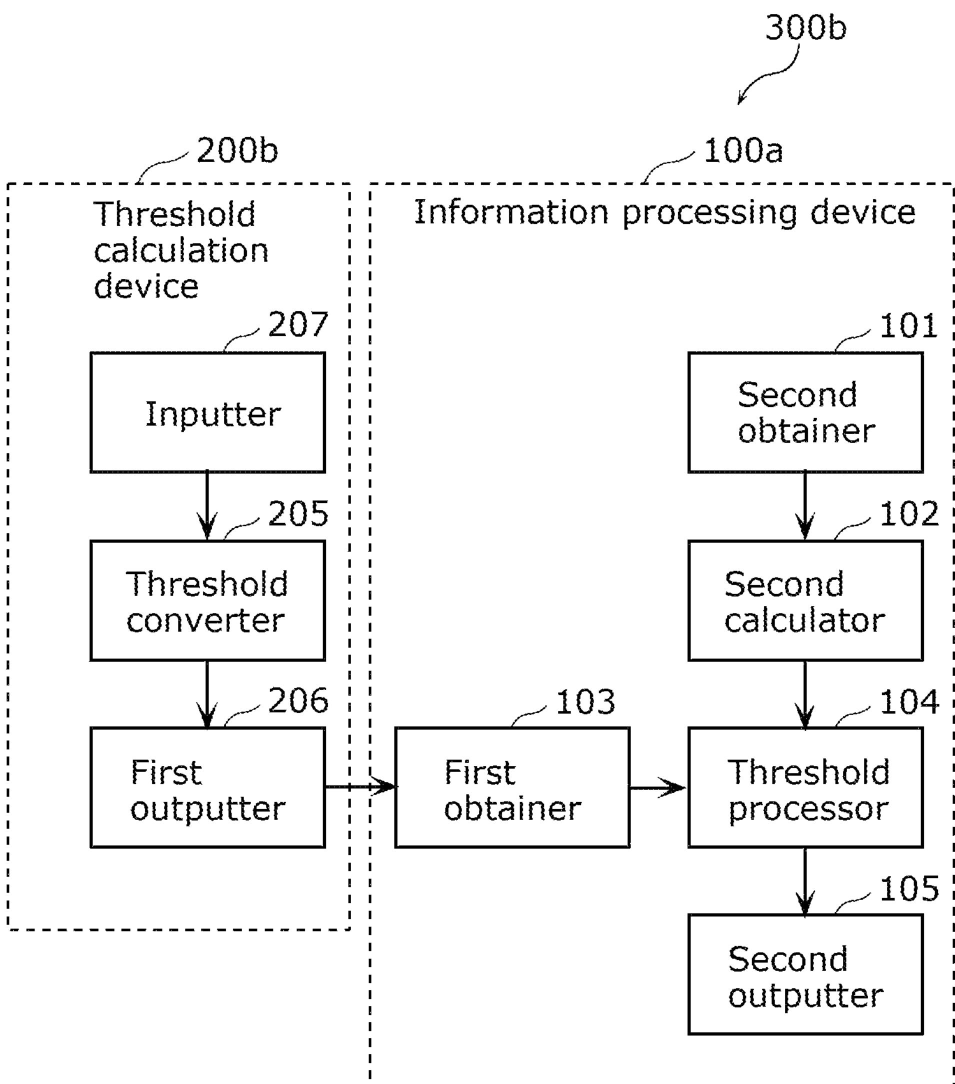

300b
200b
Threshold calculation device
207
Inputter
205
Threshold converter
206
First outputter
100a
Information processing device
101
Second obtainer
102
Second calculator
103
First obtainer
104
Threshold processor
105
Second outputter

FIG. 8

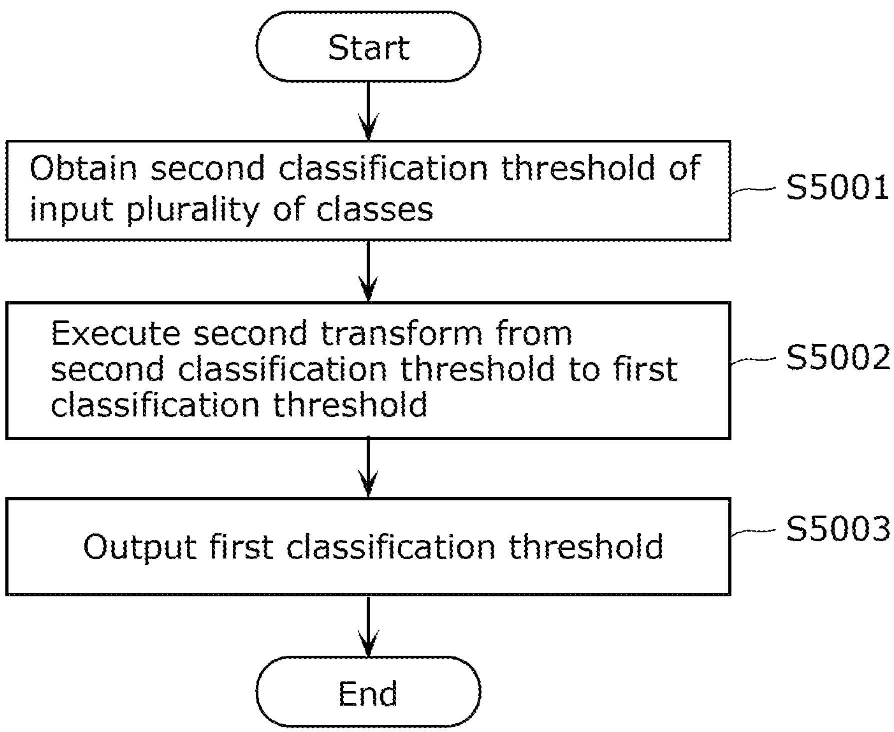

Start
Obtain second classification threshold of input plurality of classes
S5001
Execute second transform from second classification threshold to first classification threshold
S5002
Output first classification threshold
S5003
End

FIG. 9

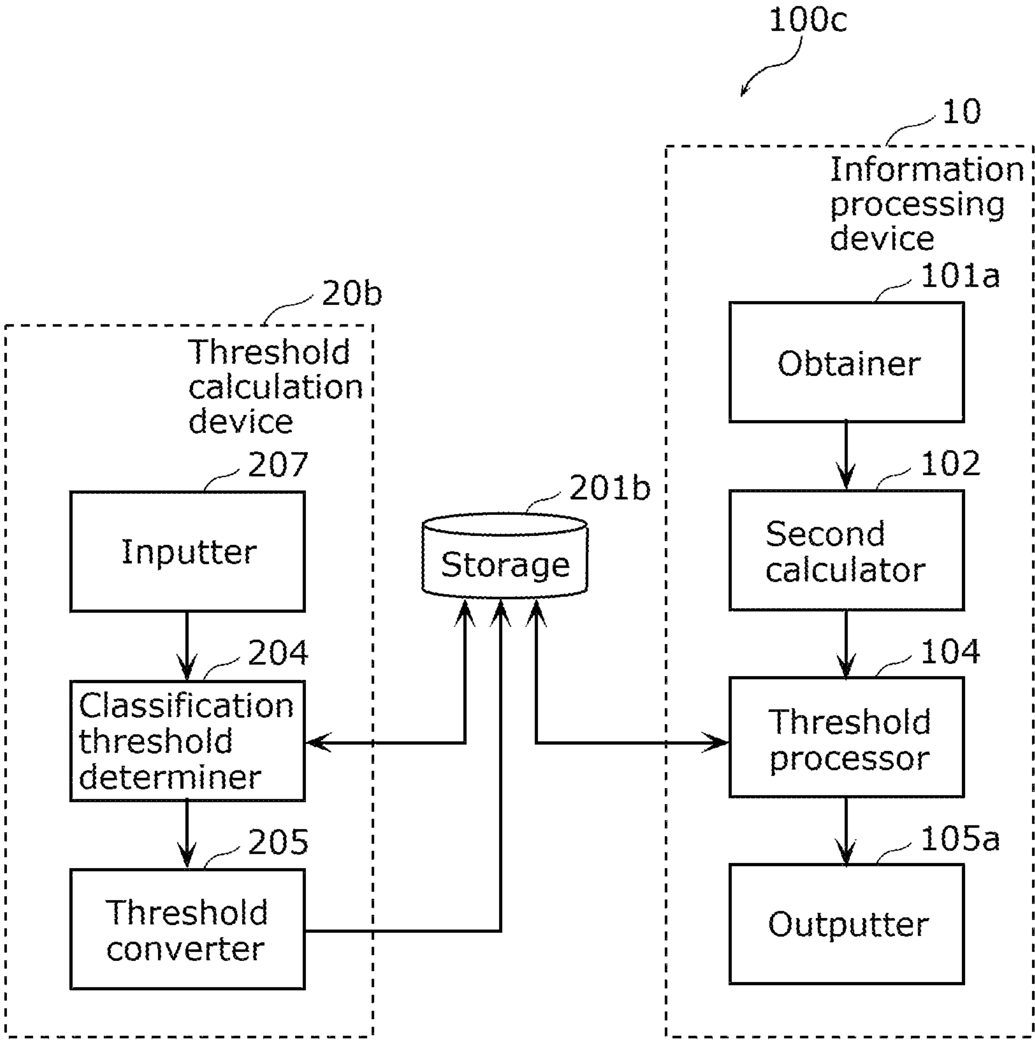
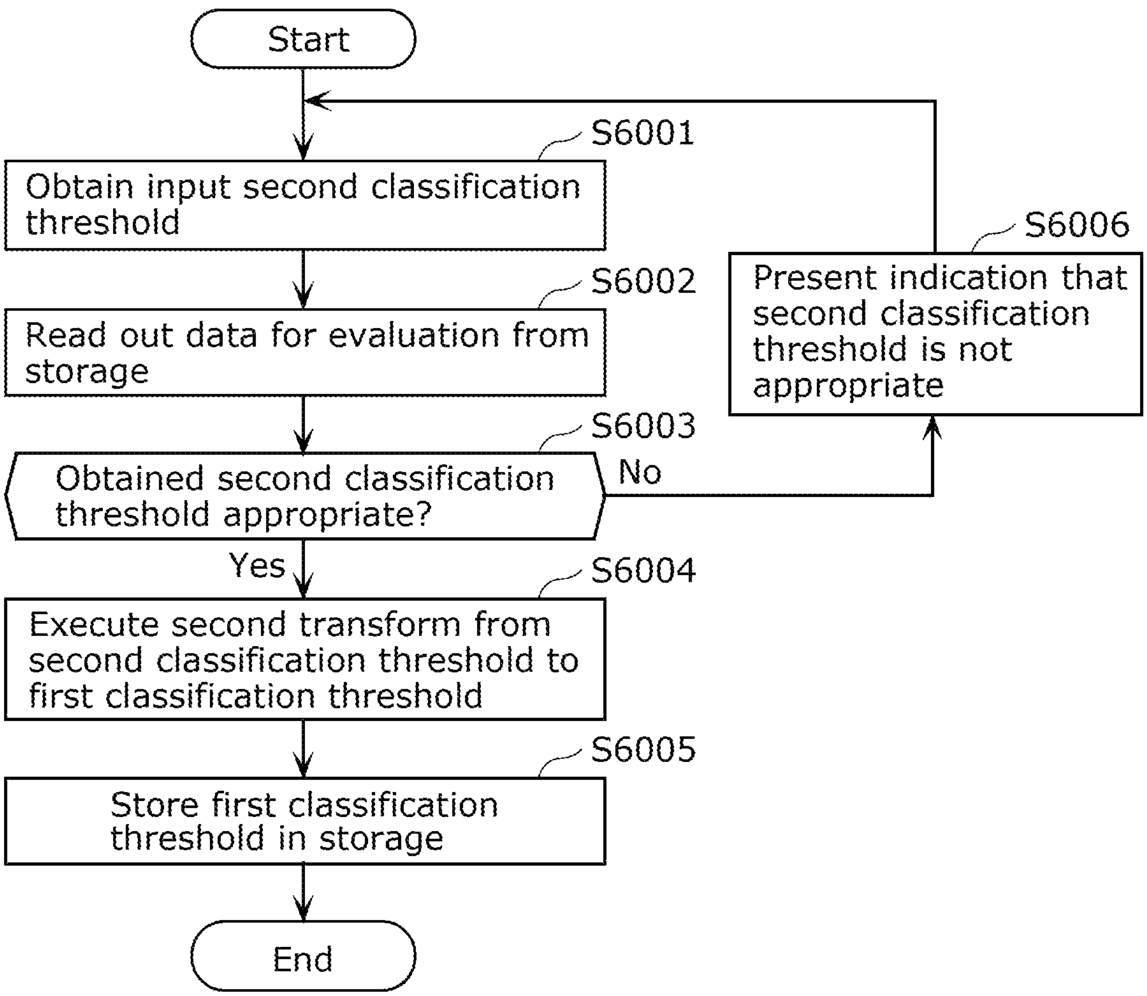

100c
10
Information processing device
101a
Obtainer
102
Second calculator
104
Threshold processor
105a
Outputter
20b
Threshold calculation device
207
Inputter
204
Classification threshold determiner
205
Threshold converter
201b
Storage

FIG. 10


Start
S6001
Obtain input second classification threshold
S6002
Read out data for evaluation from storage
S6003
Obtained second classification threshold appropriate?
No
Yes
S6004
Execute second transform from second classification threshold to first classification threshold
S6005
Store first classification threshold in storage
End
S6006
Present indication that second classification threshold is not appropriate

INFORMATION PROCESSING DEVICE, INFORMATION PROCESSING METHOD, AND RECORDING MEDIUM

CROSS REFERENCE TO RELATED APPLICATIONS

This is a continuation application of PCT International Application No. PCT/JP2019/048321 filed on Dec. 10, 2019, designating the United States of America, which is based on and claims priority of Japanese Patent Application No. 2019-107878 filed on Jun. 10, 2019 and U.S. Provisional Patent Application No. 62/787,576 filed on Jan. 2, 2019. The entire disclosures of the above-identified applications, including the specifications, drawings and claims are incorporated herein by reference in their entirety.

FIELD

The present disclosure relates to an information processing device, an information processing method, and a recording medium.

BACKGROUND

In fields such as image recognition, multilayer neural networks (Deep Neural Networks, or "DNNs") are used as a recognition model (also called a "classification model" hereinafter) to recognize objects in images. A DNN, for example, takes an image as an input, and outputs a probability value of an object in the Image being classified into a class (also called a "likelihood" of an object class). In this case, a Softmax function is used in an output layer of the DNN (for example, see PTL 1).

CITATION LIST

Patent Literature

PTL 1: International Publication No. 2017/149722

SUMMARY

Technical Problem

However, a Softmax function involves exponential computations, which may place a strain on computational resources when the classification model is implemented on embedded devices having limited computational resources.

Accordingly, the present disclosure provides an Information processing device, an information processing method, and a recording medium capable of reducing an amount of computations for classifying an object into a class.

Solution to Problem

To solve the above-described problem, an information processing device according to one aspect of the present disclosure is an information processing device including a processor. The processor is configured to obtain a first classification threshold for classifying data into at least one of a plurality of classes, and output a classification result of classifying the data into at least one of the plurality of classes based on an output of a trained classification model and the first classification threshold. The first classification threshold is obtained by a second transform performed on a second classification threshold, the second transform being an inverse transform of a first transform. The first transform corresponds to transforming the output of the trained classification model into a classification probability value of each of a plurality of unit classes constituting the plurality of classes. The second classification threshold is set based on the classification probability values of the plurality of unit classes.

Additionally, an information processing method according to one aspect of the present disclosure is an information processing method executed by a computer. The information processing method includes: executing a first transform from an output of a trained classification model into a classification probability value of each of a plurality of unit classes; setting a second classification threshold based on the classification probability values of the plurality of unit classes; executing a second transform, the second transform being a transform from the second classification threshold into a first classification threshold for classifying data into at least one of a plurality of classes, and the second transform being an inverse transform of the first transform; and outputting the first classification threshold.

Additionally, a recording medium according to one aspect of the present disclosure is a non-transitory computer-readable recording medium having a program recorded thereon for causing a computer to execute an information processing method. The information processing method includes: obtaining a first classification threshold for classifying data into at least one of a plurality of classes; and outputting a classification result of classifying the data into at least one of the plurality of classes based on an output of a trained classification model and the first classification threshold. The first classification threshold is obtained by a second transform performed on a second classification threshold, the second transform being an inverse transform of a first transform. The first transform corresponds to transforming the output of the trained classification model into a classification probability value of each of a plurality of unit classes. The second classification threshold is set based on the classification probability values of the plurality of unit classes.

Advantageous Effects

According to the present disclosure, the amount of computations for classifying an object into a class can be reduced.

BRIEF DESCRIPTION OF DRAWINGS

These and other advantages and features will become apparent from the following description thereof taken in conjunction with the accompanying Drawings, by way of non-limiting examples of embodiments disclosed herein.

FIG. **1** is a block diagram illustrating an example of the configuration of an information processing system according to Embodiment 1.

FIG. **2** is a flowchart illustrating an example of operations of a threshold calculation device according to Embodiment 1.

FIG. **3** is a flowchart illustrating an example of operations performed by an information processing device according to Embodiment 1.

FIG. **4** is a block diagram illustrating an example of the configuration of an information processing device according to a variation on Embodiment 1.

FIG. **5** is a flowchart illustrating an example of operations of a threshold calculator according to a variation on Embodiment 1.

FIG. **6** is a flowchart illustrating an example of operations of an information processor according to a variation on Embodiment 1.

FIG. **7** is a block diagram illustrating an example of the configuration of an information processing system according to Embodiment 2.

FIG. **8** is a flowchart illustrating an example of operations of a threshold calculation device according to Embodiment 2.

FIG. **9** is a block diagram illustrating an example of the configuration of an information processing device according to a variation on Embodiment 2.

FIG. **10** is a flowchart illustrating an example of operations of a threshold calculator according to a variation on Embodiment 2.

DESCRIPTION OF EMBODIMENTS

(Underlying Knowledge Leading to Present Disclosure)

Thus far, multilayer neural networks (Deep Neural Networks, "DNNs") implemented in computing devices with limited computational resources, such as embedded devices, have had an issue in that the number of hidden layer units cannot be Increased, which causes a drop in pattern recognition performance. In response to this issue, the past technique described in PTL 1 determines whether or not to perform scalar quantization in each layer of a DNN, and in the next layer after the layer in which scalar quantization is performed, multiplies the scalar-quantized vector with a weight vector. This makes it possible to reduce the amount of computations more than when multiplying a non-scalar-quantized vector and the weight vector, which in turn makes it possible to increase the number of hidden layer units. However, a likelihood vector is calculated using a scalar-quantized output vector, and thus the values are coarser than when calculating the likelihood vector using a non-scalar-quantized output vector. The recognition accuracy may drop as a result.

In the past technique described in PTL 1, when scalar quantization has not been performed in the layer one previous to the output layer of the DNN, a Softmax function is applied in the output layer to calculate a likelihood vector ("classification probability value" hereinafter) for a plurality of classes. The Softmax function involves the computation of exponential functions, and thus the amount of computations of exponential functions becomes an issue when implemented in embedded systems. Furthermore, a Softmax function adjusts a sum to be 1 with respect to the input, and thus original values cannot be restored even through Inverse transforms. In other words, because the Softmax function performs an irreversible computation on the input, a non-normalized value cannot be obtained by performing an inverse transform on the output value obtained in response to the input to the Softmax function. Therefore, in the past technique described in PTL 1, it is necessary, for example, to calculate a classification probability value for each of a plurality of classes of objects in an input Image in order to identify an object in the input image. Such a process of calculating the classification probability value of an object in an input image increases the amount of computations in computing devices with limited computational resources, such as embedded devices, and may therefore reduce the recognition accuracy of DNNs implemented in such computing devices. It is therefore difficult to say that the past technique described in PTL 1 is able to reduce the amount of computations for classifying objects into classes.

After diligently examining the above-described issue, the inventors of the present disclosure found that, in the process of determining a threshold for each of a plurality of classes, performing a reversible transform at the output layer of the DNN makes it possible to perform an inverse transform on the calculated threshold and obtain a non-normalized threshold. The inventors therefore arrived at an information processing device that, for example, can use a non-normalized threshold in a process of classifying objects in an input image into classes, which makes it possible to reduce the amount of computations for classifying objects into classes.

An overview of one aspect of the present disclosure is as follows.

An information processing device according to one aspect of the present disclosure is an information processing device including a processor. The processor is configured to obtain a first classification threshold for classifying data into at least one of a plurality of classes, and output a classification result of classifying the data into at least one of the plurality of classes based on an output of a trained classification model and the first classification threshold. The first classification threshold is obtained by a second transform performed on a second classification threshold, the second transform being an inverse transform of a first transform. The first transform corresponds to transforming the output of the trained classification model into a classification probability value of each of a plurality of unit classes constituting the plurality of classes. The second classification threshold is set based on the classification probability values of the plurality of unit classes.

According to the above-described configuration, an inverse-transformable function is used in the first transform the transforms from the output of the classification model into the classification probability values of the plurality of unit classes. Accordingly, when the first classification threshold, which is obtained by executing the second transform that is an inverse transform of the first transform on the second classification threshold, is used in processing for classifying an object into a class, it is no longer necessary to convert the output of the classification model, which takes, for example, an image as an input, into the classification probability values of the plurality of unit classes. An information processing device according to one aspect of the present disclosure can therefore reduce the amount of computations performed for classifying an object into a class.

Specifically, in an information processing device according to one aspect of the present disclosure, the output of the trained classification model may be a plurality of scalars corresponding to the plurality of classes.

Additionally, an information processing method according to one aspect of the present disclosure is an information processing method executed by a computer. The information processing method includes: executing a first transform from an output of a trained classification model into a classification probability value of each of a plurality of unit classes; setting a second classification threshold based on the classification probability values of the plurality of unit classes; executing a second transform, the second transform being a transform from the second classification threshold into a first classification threshold for classifying data into at least one of a plurality of classes, and the second transform being an inverse transform of the first transform; and outputting the first classification threshold.

According to the above-described method, an inverse-transformable function is used in the first transform the transforms from the output of the classification model into the classification probability values of the plurality of unit classes. As such, the first classification threshold, which is a non-normalized threshold, is obtained by executing the second transform, which is an inverse transform of the first transform, on the second classification threshold. For example, if the first classification threshold is used in processing for classifying an object into a class, it is no longer necessary to convert the output of the classification model into the classification probability values of the plurality of unit classes. Accordingly, according to the Information processing method according to one aspect of the present disclosure, the first classification threshold, which is a non-normalized threshold, is obtained, which makes it possible to reduce the amount of computations performed for classifying an object into a class.

For example, in an Information processing method according to one aspect of the present disclosure, the first transform may be a computation by an inverse-transformable probability function, and the second transform may be a computation by an inverse function of the inverse-transformable probability function.

Accordingly, by inverse-transforming the classification probability values of the plurality of unit classes, pre-transform values, i.e., the output of the trained classification model, can be derived.

For example, in an information processing method according to one aspect of the present disclosure, the first transform may be a database corresponding to a computation by an inverse-transformable probability function, and the second transform may be a database corresponding to a computation by an inverse function of the inverse-transformable probability function.

This makes it possible to further reduce the amount of calculations than when using function computations.

For example, an information processing method according to one aspect of the present disclosure may further include: obtaining a data set; obtaining the classification probability value of each of the plurality of unit classes for each of data included in the data set by inputting the data set into the trained classification model; and determining the second classification threshold based on a classification result obtained by using the second classification threshold on each obtained classification probability value of each of the plurality of unit classes.

Through this, a classification probability value threshold for the classification probability values of the plurality of unit classes, i.e., the second classification threshold, is shifted with reference to correct answer data contained in the data set for evaluation, and a second classification threshold which satisfies a target accuracy is selected. Therefore, with the information processing method according to one aspect of the present disclosure, a threshold at which a desired classification accuracy is obtained can be determined.

Additionally, a recording medium according to one aspect of the present disclosure is a non-transitory computer-readable recording medium having a program recorded thereon for causing a computer to execute an information processing method. The information processing method includes: obtaining a first classification threshold for classifying data into at least one of a plurality of classes; and outputting a classification result of classifying the data into at least one of the plurality of classes based on an output of a trained classification model and the first classification threshold. The first classification threshold is obtained by a second transform performed on a second classification threshold, the second transform being an inverse transform of a first transform. The first transform corresponds to transforming the output of the trained classification model into a classification probability value of each of a plurality of unit classes. The second classification threshold is set based on the classification probability values of the plurality of unit classes.

According to the above-described recording medium, an inverse-transformable function is used in the first transform the transforms from the output of the classification model into the classification probability values of the plurality of unit classes. Accordingly, when the first classification threshold, which is obtained by executing the second transform that is an inverse transform of the first transform on the second classification threshold, is used in processing for classifying an object into a class, it is no longer necessary to convert the output of the classification model, which takes, for example, an image as an input, into the classification probability values of the plurality of unit classes. A recording medium according to one aspect of the present disclosure can therefore reduce the amount of computations performed for classifying an object into a class.

Embodiments of the present disclosure will be described hereinafter with reference to the drawings.

Note that the following embodiments describe comprehensive or specific examples of the present disclosure. The numerical values, shapes, constituent elements, arrangements and connection states of constituent elements, steps, orders of steps, and the like in the following embodiments are merely examples, and are not intended to limit the present disclosure. Additionally, of the constituent elements in the following embodiments, constituent elements not denoted in the independent claims will be described as optional constituent elements.

Additionally, the drawings are schematic diagrams, and are not necessarily exact illustrations. As such, the scales and so on, for example, are not necessarily consistent from drawing to drawing. Furthermore, configurations that are substantially the same are given the same reference signs in the drawings, and redundant descriptions will be omitted or simplified.

Additionally, in the present specification, terms indicating relationships between elements, such as "horizontal" or "vertical", and numerical value ranges do not express the items in question in the strictest sense, but rather include substantially equivalent ranges, e.g., differences of several percent, as well.

Embodiment 1

Overview of Information Processing System

First, an overview of an information processing system including an information processing device according to Embodiment 1 will be described with reference to the drawings. FIG. **1** is a block diagram illustrating an example of the configuration of information processing system **300***a* according to Embodiment 1.

Information processing system **300***a* is a system that classifies data obtained by a sensor into at least one of a plurality of classes and outputs a classification result. Information processing system **300***a* includes threshold calculation device **200***a*, which calculates a first classification threshold for classifying the data into at least one of the plurality of classes, and Information processing device **100***a*, which based on an output of a trained classification model and the first classification threshold, outputs a classification result in which the data has been classified into at least one of the plurality of classes.

The sensor is, for example, a sound sensor such as a microphone, an image sensor, a range sensor, a gyrosensor, a pressure sensor, or the like. Data obtained using a plurality of sensors may be obtained using a three-dimensional reconstruction technique such as SfM (Structure from Motion), for example. The data obtained by the sensor is, for example, audio, an image, a moving image, three-dimensional point cloud data, or vector data.

In Information processing system **300***a*, a plurality of classes that classify the data may be set in accordance with the type of the data, the application of the data, or the like. For example, if the data is audio, classes such as the voice of a specific person, the operation sound of a specific machine, or the cry of a specific animal may be set. If the data is an image, in a surveillance camera system, for example, a class such as a specific person may be set, and in an in-vehicle camera system, for example, a class such as pedestrian, automobile, motorcycle, bicycle, background, and the like may be set. If the data is three-dimensional point cloud data, for example, a class such as unevenness or cracks in a structure or terrain, or a specific structure, may be set from the three-dimensional shape of a structure or terrain. Finally, if the data is vector data, for example, a class such as motion vectors at a plurality of parts of a structure, such as bridge girders or soundproof walls, may be set.

Each element of information processing system **300***a* will be described below.

Threshold Calculation Device

Threshold calculation device **200***a* is a device for calculating a first classification threshold for classifying data into at least one of a plurality of classes.

As illustrated in FIG. **1**, threshold calculation device **200***a* includes storage **201**, first calculator **202**, classification probability calculator **203**, classification threshold determiner **204**, threshold converter **205**, and first outputter **206**.

Storage **201** stores a data set for evaluation of a second classification threshold. The data set for evaluation includes a set of input data to be input to first calculator **202** and correct answer data corresponding to the input data. The correct answer data are a classification probability value for each of a plurality of classes of the input data. Hereinafter, the classification probability values for each of the plurality of classes will also be referred to as classification probability values for a plurality of unit classes. “Unit class” refers to each class constituting the plurality of classes. The classification probability value of a plurality of unit classes is a normalized probability value obtained by classification probability calculator **203** performing a first transform on an output of first calculator **202**. The second classification threshold is set based on the classification probability values of the plurality of unit classes.

First calculator **202** is a feature amount extractor that extracts a feature amount of the data, e.g., a machine learning model. For example, first calculator **202** is a trained classification model. The classification model is a multilayer neural network (DNN). First calculator **202** obtains the data set for evaluation. For example, first calculator **202** reads out the data set for evaluation from storage **201**. The input data of the data set for evaluation is input to first calculator **202**. First calculator **202** outputs a plurality of scalars corresponding to the plurality of classes of the input data. Each scalar is a feature amount that corresponds to a respective class of the input data. The output of first calculator **202** is a non-normalized value.

Note that first calculator **202** is not limited to a DNN. For example, first calculator **202** may be a feature amount extractor, aside from a DNN, that uses a method such as edge extraction, primary component analysis, block matching, or a sampling moiré method.

Note that first calculator **202** may obtain the data set for evaluation from another device through communication. For example, the data set for evaluation may be obtained from a server or storage over the Internet.

Classification probability calculator **203** performs the first transform, which is a transform of the output of first calculator **202** to classification probability values of the plurality of unit classes. More specifically, in the first transform, classification probability calculator **203** calculates the classification probability values of the plurality of unit classes from the output of first calculator **202** using a reversible transform. For example, classification probability calculator **203** derives classification probability values corresponding to the plurality of classes of input data by normalizing a plurality of scalars (feature amounts) corresponding to the plurality of classes of input data using a probability function that can be inverse-transformed. By performing normalization using a reversible transform in this manner, a non-normalized threshold can be derived by inverse-transforming a suitable threshold after determining the suitable threshold (the second classification threshold; described later) based on the data set for evaluation.

Classification probability calculator **203** is constituted by, for example, probability calculators of a plurality of unit classes. Each of the plurality of scalars corresponding to the plurality of classes is input to the probability calculator of the corresponding unit class among the plurality of classes. Each of the probability calculators of the plurality of unit classes is an inverse-transformable function. These functions may be different from each other, or may be the same. The inverse-transformable function may be a differentiable function, e.g., a sigmoid function, a tangent hyperbolic function (Tan h), or the like. Note that the first transform may be a computation performed using an inverse-transformable probability function, or may be a transform performed using a database corresponding to a computation performed using an inverse-transformable probability function. The database may be a table in which inputs and outputs (post-transform values) are mapped to each other, such as a lookup table, for example.

Classification threshold determiner **204** determines the second classification threshold based on the classification probability values of the plurality of unit classes, which have been calculated by classification probability calculator **203**. To be more specific, classification threshold determiner **204** obtains the classification probability values of the plurality of unit classes, and determines the second classification threshold based on a classification result obtained using the second classification threshold for each of the obtained classification probability values of the plurality of unit classes. For example, classification threshold determiner **204** reads out the data set for evaluation from storage **201**, and determines the second classification threshold based on the correct answer data in the data set for evaluation and the classification probability values of the plurality of unit classes which have been calculated by classification probability calculator **203**. In other words, classification threshold determiner **204** determines an optimal second classification threshold based on the data set for evaluation. For example, classification threshold determiner **204** determines the second classification threshold in accordance with a false positive (FP)/false negative (FN) ratio with respect to the data set for evaluation, for each classification probability value of the plurality of unit classes.

Note that the second classification threshold may be set to a predetermined value, or may be determined in accordance with a target accuracy set by a user. When the second classification threshold is determined in accordance with the target accuracy, classification threshold determiner **204** may determine the second classification threshold so that a result obtained by applying the second classification threshold to the classification probability values of the plurality of unit classes satisfies a target threshold. Note that the target accuracy may be set on a class-by-class basis, or may be set to be common across all classes. This method will be described in detail later in the section pertaining to threshold calculation device operations.

Note that the second classification threshold may be a different value for each of the plurality of unit classes, or may be the same value for all the plurality of unit classes.

Threshold converter **205** performs a second transform, which is a transform from the second classification threshold into the first classification threshold for classifying the data into at least one of a plurality of classes, and is an inverse transform of the first transform. In other words, threshold converter **205** converts the second classification threshold into a non-normalized threshold (i.e., the first classification threshold) by performing an inverse transform. Threshold converter **205** may be an inverse function of a function constituting classification probability calculator **203** (e.g., an inverse-transformable probability function), or may be a database corresponding to computations made using an inverse function of an inverse-transformable probability function. The database may be a table in which inputs and outputs (post-inverse transform values) are mapped to each other, such as a lookup table, for example.

The first classification threshold is used in information processing device **100***a* for classifying the data into at least one of the plurality of classes. Information processing device **100***a* can execute the processing of classifying the data into classes by using a non-normalized threshold (the first classification threshold). Accordingly, with information processing device **100***a*, the classification process can be executed based on feature amounts extracted from the data, which eliminates the need for normalization processing and makes it possible to reduce the amount of computations.

Note that the first classification threshold may be set to a different value for each of the plurality of classes, or may be set to a value which is common for all of the plurality of classes.

First outputter **206** outputs the first classification threshold. More specifically, first outputter **206** outputs the first classification threshold to first obtainer **103** of information processing device **100***a* through communication. The communication may be wireless communication such as Wi-Fi (registered trademark) or Bluetooth (registered trademark), or wired communication such as Ethernet (registered trademark), for example.

Note that threshold calculation device **200***a* may include a trainer (not shown) for training a machine learning model. The trainer may include storage (not shown) that holds a data set for training. The data set for training includes a set of input data and correct answer data, stored in advance. The trainer may update the data set for training by obtaining new data for training from a database located on the server connected over a communication network such as the Internet. Additionally, the trainer may include a holder (not shown) that holds the same classification model as first calculator **202**, and the holder may further hold the same N single-class classification probability calculators as classification probability calculator **203**. The trainer trains the same classification model as first calculator **202** by using the data set for training. Furthermore, the trainer may train a network having N single-class classification probability calculators provided by classification probability calculator **203**. Once the training of the classification model and the network ends, the trainer may output the trained classification model to first calculator **202**, and update first calculator **202** to the trained classification model. Likewise, the trainer may output the trained network having the N single-class classification probability calculators to classification probability calculator **203**, and update classification probability calculator **203** to the trained network.

Information Processing Device

Information processing device **100***a* will be described next. Information processing device **100***a* is a device that, for example, classifies data obtained by a sensor into at least one of a plurality of classes and outputs a classification result. The following will describe an example in which the data is an image. Note that the data, and the sensor that obtains the data, have already been described in the overview of information processing system **300***a*, and will therefore not be described here.

As illustrated in FIG. **1**, information processing device **100***a* includes first obtainer **103**, second obtainer **101**, second calculator **102**, threshold processor **104**, and second outputter **105**.

First obtainer **103** obtains the first classification threshold output from first outputter **206** of threshold calculation device **200***a*, and outputs the obtained first classification threshold to threshold processor **104**, through communication. The communication has already been described above, and will therefore not be mentioned here. Note that the first classification threshold may be stored in advance in storage included in information processing device **100***a*.

Second obtainer **101** obtains the data from the sensor through communication. Here, the sensor is an image sensor, and the data is an image. Second obtainer **101** outputs the obtained data to second calculator **102**. Note that the communication may be wireless communication or wired communication. Additionally, the sensor is not limited to a single sensor, and the data may be obtained by synchronizing two or more sensors, for example. Additionally, second obtainer **101** may obtain the data through removable memory. The removable memory is, for example, USB (Universal Serial Bus) memory.

Second calculator **102** is a feature amount extractor that extracts a feature amount of the data, e.g., a machine learning model. For example, second calculator **102** is a trained classification model. The classification model is a multilayer neural network (DNN). The data obtained by first obtainer **103** is input to second calculator **102**. Second calculator **102** outputs a plurality of scalars corresponding to the plurality of classes of the data. Each scalar is a feature amount that corresponds to a respective class of the data. The output of second calculator **102** is a non-normalized value.

Note that second calculator **102** is not limited to a DNN. For example, second calculator **102** may be a feature amount extractor, aside from a DNN, that uses a method such as feature point extraction (e.g., edge extraction), primary component analysis, block matching, or a sampling moiré method.

Threshold processor **104** obtains the first classification threshold output from first obtainer **103**, and classifies the data into at least one of a plurality of classes based on the output of second calculator **102** and the first classification threshold. To be more specific, threshold processor **104** classifies the data into at least one of the plurality of classes by determining whether or not a probability value for each of the plurality of classes, output from second calculator **102**, is at least the first classification threshold. The first classification threshold may be set to a different value for each of the plurality of classes, or may be set to a value which is common for all of the plurality of classes. The specific operations performed by threshold processor **104** will be described later.

Second outputter **105** outputs a data classification result. Second outputter **105** may output the data classification result to a presenter (not shown), or to another device aside from information processing device **100***a*. For example, second outputter **105** may cause the presenter to present information based on the classification result, on the basis of a user operation input to an inputter (not shown). The inputter is, for example, a keyboard, a mouse, a touch panel, a button, a microphone, or the like. The presenter is, for example, a display, a speaker, or the like. Note that information processing device **100***a* may or may not include the inputter and the presenter. The inputter and the presenter may be provided in another device aside from information processing device **100***a*, for example. The other device aside from information processing device **100***a* may be, for example, an information terminal such as a smartphone, a tablet, a computer, or the like. Additionally, although information processing device **100***a* has been described as a computer as an example, information processing device **100***a* may be provided in a server connected over a communication network such as the Internet.

Note that like threshold calculation device **200***a*, information processing device **100***a* may include a trainer (not shown) for training a machine learning model. With respect to the details of the trainer, the same descriptions as those given of the trainer of threshold calculation device **200***a* may be applied, with the exception that training of the network of classification probability calculator **203** is not included, and first calculator **202** is replaced with second calculator **102**.

Note that information processing system **300***a* may include a trainer (not shown) that is shared by threshold calculation device **200***a* and information processing device **100***a*, and that trainer may train the classification model of first calculator **202**, the network of classification probability calculator **203**, and the classification model of second calculator **102**.

Operations of Threshold Calculation Device

Operations performed by threshold calculation device **200***a* will be described next with reference to FIG. **2**. FIG. **2** is a flowchart illustrating an example of operations of threshold calculation device **200***a* according to Embodiment 1.

First calculator **202** reads out the input data of the data set for evaluation from storage **201**, and outputs a plurality of scalars corresponding to the plurality of classes of the input data (step S**1001**). The plurality of scalars corresponding to the plurality of classes of the input data are feature amounts corresponding to each of the plurality of classes of the input data. Assume, for example, that the input data is an image, and an automobile and a motorcycle appear in the image. Assume also that the plurality of classes are, for example, "pedestrian", "automobile", "motorcycle", "bicycle", and "background". At this time, first calculator **202** outputs a vector having a plurality of scalars corresponding to the plurality of classes in the input data (pedestrian, automobile, motorcycle, bicycle, background)=(0.1, 90, 60, 0.01, 0.001).

Next, classification probability calculator **203** performs the first transform, which is a transform of the output of first calculator **202** to classification probability values of the plurality of unit classes (step S**1002**). As described above, classification probability calculator **203** includes classification probability calculators of a plurality of unit classes. The classification probability calculator of each unit class is an inverse-transformable probability function. The classification probability calculators of the unit classes may be mutually-different functions, or may be the same function. Additionally, the classification probability calculators of the unit classes may be a database in which those calculators are associated with computations performed by inverse-transformable probability functions. A threshold calculator in each unit class converts the scalar for each of the plurality of classes into a value in a range from 0 to 1 through a reversible transform (i.e., normalizes the scalar). For example, when the scalars of (pedestrian, automobile, motorcycle, bicycle, background)=(0.1, 90, 60, 0.01, 0.001), which is the output of first calculator **202**, are input to the classification probability calculators of the unit classes corresponding to each class, the classification probability calculator of each unit class converts that scalar into a value in a range from 0 to 1. Classification probability calculator **203** outputs (pedestrian, automobile, motorcycle, bicycle, background)=(0.3, 1.0, 1.0, 0.1, 0), which are the classification probability values of the plurality of unit classes. Here, "normalization" refers not to adding a plurality of scalar values to adjust the values to 1, but rather to converting the scalar values to values in a range from 0 to 1 in accordance with the magnitudes of those scalar values. The transform coefficient used at this time may be adjusted in accordance with the classification accuracy of each class.

Next, classification threshold determiner **204** sets the second classification threshold based on the classification probability values of the plurality of unit classes, which have been derived in step S**1002** (step S**1003**). To be more specific, classification threshold determiner **204** obtains the classification probability values of the plurality of unit classes, and determines the second classification threshold based on a classification result obtained using the second classification threshold for each of the obtained classification probability values of the plurality of unit classes. For example, classification threshold determiner **204** reads out the data set for evaluation from storage **201**, and determines the second classification threshold based on the correct answer data in the data set for evaluation and the classification probability values of the plurality of unit classes. For example, classification threshold determiner **204** may determine the second classification threshold in accordance with a false positive (FP)/false negative (FN) ratio with respect to the data set for evaluation, for each classification probability value of the plurality of unit classes. Additionally, for example, the second classification threshold may be determined so that a result obtained by applying the second classification threshold to the outputs of the plurality of unit classes (i.e., the classification probability values) satisfies a target accuracy. To be more specific, a target accuracy may be provided; classification accuracies for when the threshold of the classification probability values of the plurality of unit classes (the second classification threshold) is shifted may be calculated; a threshold at which the calculated classification accuracy is closest to the target accuracy may be selected; and that threshold may be determined as the second classification threshold. Alternatively, for example, the first threshold at which the calculated classification accuracy exceeds the target accuracy may be determined as the second classification threshold, instead of the threshold at which the calculated classification accuracy is closest to the target accuracy. The target accuracy may, for example, be set on a class-by-class basis, or may be set to be common across all classes. Note that the second classification threshold may be a different value for each of the plurality of unit classes, or may be the same value for all the plurality of unit classes.

Next, threshold converter **205** performs a second transform, which is a transform from the second classification threshold set by classification threshold determiner **204** into the first classification threshold for classifying the data into at least one of a plurality of classes, and is an inverse transform of the first transform (step S**1004**). Through this, a non-normalized threshold (here, the first classification threshold) can be obtained from a normalized threshold (here, the second classification threshold). The first classification threshold is a threshold for classifying the data into at least one of a plurality of classes. The first classification threshold may be set to a different value for each of the classes, or may be set to a value which is common for all of the plurality of classes. For example, if threshold converter **205** is an inverse function of the function constituting classification probability calculator **203** (e.g., an inverse-transformable probability function), threshold converter **205** inputs the second classification threshold derived in step S**1003** and calculates the first classification threshold by inverse-transforming the second classification threshold. Additionally, if threshold converter **205** is a database corresponding to computations performed by an inverse function of the function constituting classification probability calculator **203** (e.g., a table in which inputs and outputs are mapped to each other, such as a lookup table), when the second classification threshold is input to threshold converter **205**, the first classification threshold associated with the input is output.

Next, first outputter **206** outputs the first classification threshold derived in step S**1004** to information processing device **100***a* (step S**1005**). At this time, first outputter **206** may be communicably connected to information processing device **100***a*. The communication method has already been described above, and will therefore not be mentioned here.

Operations of Information Processing Device

Operations performed by information processing device **100***a* will be described next with reference to FIG. **3**. FIG. **3** is a flowchart illustrating an example of operations performed by information processing device **100***a* according to Embodiment 1.

Second obtainer **101** obtains data from, for example, a sensor such as an image sensor (not shown) (step S**2001**). An example in which the data is in image will be described here. The image may be a moving image, or may be a still image (also called simply an "image"). Second obtainer **101** may be communicably connected to the sensor, or may obtain a plurality of images from the sensor through removable memory, e.g., USB (Universal Serial Bus) memory. The communication method has already been described above, and will therefore not be mentioned here.

Second calculator **102** outputs a plurality of scalars corresponding to the plurality of classes of the data obtained in step S**2001** (step S**2002**). The data may be, for example, a moving image captured by an in-vehicle camera, or may be an image. When the data is a moving image, the following operations are executed for each of a plurality of frames constituting the moving image. Like the operation of first calculator **202** of threshold calculation device **200***a*, second calculator **102** outputs a plurality of scalars (feature amounts) from the obtained data. The following will describe an example in which the output of second calculator **102** is (pedestrian, automobile, motorcycle, bicycle, background)=(70, 90, 0.5, 60, 0.001).

First obtainer **103** obtains the first classification threshold output from threshold calculation device **200***a* (step S**2003**). At this time, first obtainer **103** obtains the first classification threshold from threshold calculation device **200***a* through communication. The first classification threshold and the communication method have already been described above, and will therefore not be mentioned here. It is assumed here that the first classification threshold is (pedestrian, automobile, motorcycle, bicycle, background)=(60, 60, 60, 60, 60).

Threshold processor **104** classifies the data obtained by second obtainer **101** into at least one of the plurality of classes based on the output of second calculator **102** and the first classification threshold obtained by first obtainer **103** (step S**2004**). To be more specific, in step S**2004**, threshold processor **104** classifies the data into at least one of the plurality of classes by determining whether or not each of the plurality of scalars corresponding to the plurality of classes in the data is greater than or equal to the first classification threshold.

For example, when the output of second calculator **102** is (pedestrian, automobile, motorcycle, bicycle, background)=(70, 90, 0.5, 60, 0.001), and the first classification threshold for each class is (pedestrian, automobile, motorcycle, bicycle, background)=(60, 60, 60, 60, 60), the scalar values for three of the classes are at least the first classification threshold. At this time, threshold processor **104** may classify the image into the three classes, i.e., pedestrian, automobile, and bicycle, or into a single class. In the latter case, of probability values corresponding to the three classes of the image, threshold processor **104** may select the class having the highest scalar value and classify that image into a single class. In this case, of the scalar values for the aforementioned three classes, the probability value is highest for the automobile class, and thus threshold processor **104** classifies that image into the automobile class. However, of the scalar values for the three classes, threshold processor **104** may select a class having a scalar value which has the greatest difference from the first classification threshold, and may classify the image into a single class. In this case, threshold processor **104** may classify that image into the automobile class. Thus as described above, the method for classifying the data may be set as appropriate in accordance with the type of the data and the purpose of the classification.

Second outputter **105** outputs the classification result obtained in step S**2004** (step S**2005**). Second outputter **105** may output the classification result to a presenter (not shown), or to another device aside from information processing device **100***a*. For example, second outputter **105** may cause the presenter to present information based on the classification result, on the basis of a user operation input to an inputter (not shown). The inputter, the presenter, the other device, and the like have been described above, and will therefore not be mentioned here.

With respect to information based on the classification result, the classification result may be presented in a variety of formats based on settings input to the inputter. For example, if the data is an image captured by an in-vehicle camera, the information based on the classification result may be a type and number of detections of an object class detected by information processing device **100***a*, a recognition accuracy for each of object classes, a change in recognition accuracy for each of object classes depending on the weather or time of day, trends of images having low recognition accuracy, advice for avoiding danger, or the like. Information processing device **100***a* may, for example, transmit an analysis result to a database located in a server connected over the Internet, and obtain information based on the analysis result.

Variation

An information processing device according to a variation on Embodiment 1 will be described next. The following descriptions will focus on the differences from Embodiment 1, and descriptions of common points will be omitted or simplified.

Overview of Information Processing Device

FIG. **4** is a block diagram illustrating an example of the configuration of information processing device **100***b* according to the variation on Embodiment 1. Information processing device **100***b* includes threshold calculator **20***a*, storage **201***a*, and information processor **10**.

Embodiment 1 described an example in which information processing device **100***a* obtains the first classification threshold from threshold calculation device **200***a* and classifies data into at least one of a plurality of classes using the obtained first classification threshold. The main difference between information processing device **100***b* and information processing device **100***a* according to Embodiment 1 is that information processing device **100***b* includes threshold calculator **20***a*. Note that in FIGS. **1** and **4**, constituent elements which are substantially identical are given the same reference signs.

As illustrated in FIG. **1**, in Embodiment 1, threshold calculation device **200***a* outputs the first classification threshold derived by threshold converter **205** to first outputter **206**, and the first classification threshold is output to information processing device **100***a* via first outputter **206**. In the present variation, as illustrated in FIG. **4**, threshold calculator **20***a* stores the first classification threshold derived by threshold converter **205** in storage **201***a*. Then, information processor **10** reads out the first classification threshold stored in storage **201***a*.

As illustrated in FIGS. **1** and **4**, obtainer **101***a* illustrated in FIG. **4** corresponds to second obtainer **101** illustrated in FIG. **1**, and outputter **105***a* illustrated in FIG. **4** corresponds to second outputter **105** illustrated in FIG. **1**. In other words, like second obtainer **101**, obtainer **101***a* obtains data from a sensor such as an image sensor. Additionally, like second outputter **105**, outputter **105***a* outputs a classification result of classifying the data obtained by obtainer **101***a* into at least one of a plurality of classes.

Operations of Threshold Calculator

FIG. **5** is a flowchart illustrating an example of operations of threshold calculator **20***a* according to a variation on Embodiment 1. Steps S**3001** to S**3004** in FIG. **5** correspond to steps S**1001** to S**1004** in FIG. **2**, respectively, and the descriptions thereof will therefore be simplified here. Note that the operations of threshold calculator **20***a* according to the variation differ from the operations of threshold calculation device **200***a* according to Embodiment 1 in that the first classification threshold is stored in storage **201***a*.

First calculator **202** reads out the input data of the data set for evaluation from storage **201**, and outputs a plurality of scalars corresponding to the plurality of classes of the input data (step S**3001**).

Next, classification probability calculator **203** performs the first transform, which is a transform of the output of first calculator **202** to classification probability values of the plurality of unit classes (step S**3002**).

Next, classification threshold determiner **204** sets the second classification threshold based on the classification probability values of the plurality of unit classes, which have been derived in step S**3002** (step S**3003**).

Next, threshold converter **205** performs a second transform on the second classification threshold, which is a transform from the second classification threshold set by classification threshold determiner **204** into the first classification threshold for classifying the data into at least one of a plurality of classes, and is an inverse transform of the first transform (step S**3004**).

Next, threshold converter **205** stores the first classification threshold in storage **201***a* (step S**3005**).

Operations of Information Processor

FIG. **6** is a flowchart illustrating an example of operations of information processor **10** according to the variation on Embodiment 1. Steps S**4001** and S**4002** in FIG. **6** correspond to steps S**2001** and S**2002** in FIG. **3**, respectively, and the descriptions thereof will therefore be simplified here.

Obtainer **101***a* obtains data from, for example, a sensor such as an image sensor (step S**4001**). Next, second calculator **102** outputs a plurality of scalars (probability values) corresponding to the plurality of classes of the data obtained in step S**4001** (step S**4002**).

A per-class loop process is then started for each of the plurality of classes of the data.

Threshold processor **104** reads out the first classification threshold stored in storage **201***a* (step S**4003**). Although an example in which threshold processor **104** reads out the first classification threshold for each of the plurality of classes for each classification process of each class will be described here, threshold processor **104** may read out the first classification values for all of the plurality of classes at once. Note that the first classification threshold may be a different value for each of the plurality of unit classes, or may be the same value.

Threshold processor **104** classifies the data obtained by second obtainer **101** into at least one of the plurality of classes based on the output of second calculator **102** and the first classification threshold read out from storage **201***a*. First, threshold processor **104** reads out the first classification threshold of one of the plurality of classes (e.g., a pedestrian class) from storage **201***a* (step S**4003**), and then determines whether or not the scalar value of that class is at least the first classification threshold read out from storage **201***a* (step S**4004**). If the scalar value of that class is at least the first classification threshold (Yes in step S**4004**), threshold processor **104** saves the scalar value of that class in association with a number indicating the class in storage **201***a* (step S**4005**).

On the other hand, if the scalar value of the class is less than the first classification threshold (No in step S**4004**), threshold processor **104** reads out the first classification threshold of another class (e.g., an automobile class) in the plurality of classes from storage **201***a* (step S**4003**). Threshold processor **104** then determines whether or not the scalar value of that class is at least the first classification threshold read out from storage **201***a* (step S**4004**). If the scalar value of that class is at least the first classification threshold (Yes in step S**4004**), threshold processor **104** stores the scalar value of that class in association with a number indicating the class in storage **201***a* (step S**4005**).

On the other hand, if the scalar value of the class is less than the first classification threshold (No in step S**4004**), threshold processor **104** reads out another class (e.g., a motorcycle class) in the plurality of classes from storage **201***a* (step S**4003**), and executes the determination processing of step S**4004**. By repeating the same processing in this manner, the classification processing can be executed on a class-by-class basis for a plurality of scalars corresponding to a plurality of classes of data. Once the per-class loop processing ends, threshold processor **104** determines whether or not the number of classes stored in storage **201***a* is at least 1 (step S**4006**). If the number of classes stored in storage **201***a* is at least 1 (Yes in step S**4006**), threshold processor **104** outputs, to outputter **105***a*, the class number having the highest scalar value (step S**4007**). In this manner, based on the class number, outputter **105***a* outputs (not shown) a classification result (e.g., that the data indicates an automobile) In which the data has been classified into at least one of a plurality of classes.

On the other hand, if the number of classes stored in storage **201***a* is 0 (No in step S**4006**), threshold processor **104** outputs, to outputter **105***a*, a number indicating "other" (step S**4008**). "Other" indicates that there is no corresponding class. At this time, the classification result output from outputter **105***a* may, for example, indicate that the data belongs to another class, or that the data belongs to a background class.

Embodiment 2

An information processing system according to Embodiment 2 will be described next. The following descriptions will focus on the differences from Embodiment 1, and descriptions of common points will be omitted or simplified.

Overview of Information Processing System

FIG. **7** is a block diagram illustrating an example of the configuration of information processing system **300***b* according to Embodiment 2. Information processing system **300***b* includes threshold calculation device **200***b* and information processing device **100***a*. The main difference between threshold calculation device **200***b* according to Embodiment 2 and threshold calculation device **200***a* according to Embodiment 1 is that the first classification threshold is derived from the second classification threshold, which has been set by a user.

Configuration of Threshold Calculation Device

The configuration of threshold calculation device **200***b* will be described next.

As illustrated in FIG. **7**, threshold calculation device **200***b* includes inputter **207**, threshold converter **205**, and first outputter **206**. Although an example in which threshold calculation device **200***b* includes inputter **207** will be described here, the configuration is not limited thereto. For example, the inputter may be provided in another device aside from threshold calculation device **200***b*. The other device is, for example, a tablet terminal, a smartphone, a computer, or the like.

Inputter **207** inputs an operation signal from a user into threshold converter **205**. Inputter **207** is, for example, a touch panel, a keyboard, a mouse, a button, a speaker, or the like. The operation signal is, for example, a signal indicating the second classification threshold for each of the plurality of classes. The user inputs the second classification threshold into threshold converter **205** through inputter **207**. Note that the second classification threshold may be a value set in advance.

Threshold converter **205** derives the first classification threshold by performing a second transform, which is an inverse transform of the first transform, on the obtained second classification threshold. First outputter **206** outputs the first classification threshold to information processing device **100***a*.

Note that information processing device **100***a* is the same as information processing device **100***a* according to Embodiment 1, and will therefore not be described here.

Operations of Threshold Calculation Device

Operations of threshold calculation device **200***b* will be described next. FIG. **8** is a flowchart illustrating an example of operations of threshold calculation device **200***b* according to Embodiment 2.

Although not illustrated here, the user inputs the second classification threshold for each of the plurality of classes to threshold converter **205** through inputter **207**.

As illustrated in FIG. **8**, threshold converter **205** obtains the second classification threshold for each of the plurality of classes, input through inputter **207** (step S**5001**). Next, threshold converter **205** performs a second transform on the second classification threshold, which is a transform from the obtained second classification threshold into the first classification threshold for classifying the data into at least one of a plurality of classes, and is an inverse transform of the first transform (step S**5002**).

First outputter **206** outputs the first classification threshold to information processing device **100***a* (step S**5003**).

Note that the operations of information processing device **100***a* are the same as in the example described in Embodiment 1, and will therefore not be described here.

Variation

An information processing device according to a variation on Embodiment 2 will be described next. The following descriptions will focus on the differences from Embodiment 2, and descriptions of common points will be omitted or simplified.

Overview of Information Processing Device

FIG. **9** is a block diagram illustrating an example of the configuration of information processing device **100***c* according to the variation on Embodiment 2. Information processing device **100***b* includes threshold calculator **20***b*, storage **201***b*, and Information processor **10**.

Embodiment 2 described an example in which Information processing device **100***a* obtains the first classification threshold from threshold calculation device **200***b* and classifies data into at least one of a plurality of classes using the obtained first classification threshold. The main difference between information processing device **100***c* according to the present variation and information processing device **100***a* according to Embodiment 2 is that information processing device **100***c* includes threshold calculator **20***b*. Additionally, the main difference between threshold calculator **20***b* according to the present variation and threshold calculation device **200***b* according to Embodiment 2 is that threshold calculator **20***b* includes classification threshold determiner **204**. Note that in FIGS. **7** and **9**, constituent elements which are substantially identical are given the same reference signs.

As illustrated in FIG. **7**, in Embodiment 2, threshold calculation device **200***b* derives the first classification threshold by using threshold converter **205** to perform an inverse transform on the second classification threshold input by the user to threshold converter **205** through inputter **207**. Then, threshold calculation device **200***b* outputs the derived first classification threshold from first outputter **206** to information processing device **100**. In the present variation, as illustrated in FIG. **9**, threshold calculator **20***b* stores the first classification threshold derived by threshold converter **205** in storage **201***b*. Then, information processor **10** reads out the first classification threshold stored in storage **201***b*, and classifies the data into at least one of a plurality of classes.

As illustrated in FIGS. **7** and **9**, obtainer **101***a* illustrated in FIG. **9** corresponds to second obtainer **101** illustrated in FIG. **7**, and outputter **105***a* illustrated in FIG. **9** corresponds to second outputter **105** illustrated in FIG. **7**. In other words, like second obtainer **101**, obtainer **101***a* obtains data from a sensor such as an image sensor. Additionally, like second outputter **105**, outputter **105***a* outputs a classification result of classifying the data obtained by obtainer **101***a* into at least one of a plurality of classes.

Additionally, as illustrated in FIGS. **1** and **9**, classification threshold determiner **204** illustrated in FIG. **9** corresponds to classification threshold determiner **204** illustrated in FIG. **1**. Classification threshold determiner **204** determines the appropriate second classification threshold by, for example, referring to a data set for evaluation stored in storage **201** or storage **201***b*.

Although threshold calculation device **200***b* according to Embodiment 2 derives the first classification threshold from the second classification threshold input by the user, threshold calculator **20***b* according to the present variation determines the second classification threshold based on the second classification threshold input by the user and the evaluation data stored in storage **201***b*. Details of the determination process will be described in the section regarding operations.

Note that information processor **10** is the same as information processor **10** according to the variation on Embodiment 1, and will therefore not be described here.

Operations of Threshold Calculator

FIG. **10** is a flowchart illustrating an example of operations of threshold calculator **20***b* according to a variation on Embodiment 2.

Although not illustrated here, the user Inputs the second classification threshold for each of the plurality of classes to classification threshold determiner **204** through inputter **207**. Through this, as illustrated in FIG. **10**, classification threshold determiner **204** obtains the input second classification threshold (step **S6001**).

Next, classification threshold determiner **204** reads out the evaluation data from storage **201***b* (step **S6002**). Based on the data set for evaluation, classification threshold determiner **204** determines whether or not the second classification threshold obtained in step **S6001** is appropriate, and more specifically, whether or not a result obtained by applying the second classification threshold to the classification probability values of a plurality of unit classes in the data set for evaluation satisfies a target accuracy (step **S6003**). If it is determined that the second classification threshold is appropriate (Yes in step **S6003**), classification threshold determiner **204** outputs the second classification threshold to threshold converter **205**. Threshold converter **205** performs a second transform on the second classification threshold, which is a transform from the obtained second classification threshold into the first classification threshold for classifying the data into at least one of a plurality of classes, and is an inverse transform of the first transform (step **S6004**). Threshold converter **205** stores the derived first classification threshold in storage **201***b* (step **S6005**).

On the other hand, if it is determined that the second classification threshold is not appropriate (No in step **S6003**), classification threshold determiner **204** causes a presenter (not shown) to present an indication that the second classification threshold is not appropriate (step **S6006**). At this time, classification threshold determiner **204** may cause the presenter to present the second classification threshold at which a result, obtained by applying the second classification threshold to the classification probability values of the plurality of unit classes in the data set for evaluation, satisfies the target accuracy.

Note that the operations of information processor **10** is the same as information processor **10** according to the variation on Embodiment 1, and will therefore not be described here.

Other Embodiments

Although one or more aspects of an information processing device and an information processing method have been described thus far on the basis of embodiments, the present disclosure is not intended to be limited to these embodiments. Variations on the present embodiment conceived by one skilled in the art and embodiments implemented by combining constituent elements from different other embodiments, for as long as they do not depart from the essential spirit thereof, fall within the scope of the present disclosure.

For example, the processing described in the foregoing embodiments may be implemented through centralized processing using a single device (system), or may be implemented through distributed processing using a plurality of devices. Additionally, a single processor or a plurality of processors may execute the above-described programs stored in the recoding medium. In other words, the processing may be centralized processing or distributed processing.

Additionally, many changes, substitutions, additions, omissions, and the like are possible for the foregoing embodiments without departing from the scope of the claims or a scope equivalent thereto.

INDUSTRIAL APPLICABILITY

The present disclosure is useful as an information processing device, an information processing method, and a recoding medium capable of reducing an amount of computations for classifying an object into a class.

The invention claimed is:

1. An information processing system comprising:

an information processing device including a first processor; and

a threshold calculating device including a second processor,

wherein the first processor is configured to:

obtain an image captured by a camera;

obtain a plurality of first classification thresholds outputted from the threshold calculating device, the plurality of first classification thresholds being a plurality of classification thresholds that are for classifying an object included in the image into at least one of a plurality of classes and correspond to the plurality of classes;

compare a plurality of first scalars and the plurality of first classification thresholds, wherein the plurality of first scalars are obtained by inputting the image into a first neural network that is a trained first classification model, wherein the plurality of first scalars correspond to the plurality of classes, wherein the plurality of first scalars are a plurality of scalars that are not transformed into classification probability values, and wherein the plurality of first classification thresholds correspond to the plurality of classes;

classify the object into, among the plurality of classes, at least one class having a corresponding first scalar that is greater than a corresponding first classification threshold; and

output, as a classification result, an information relating to the at least one class into which the object is classified,

wherein the second processor is configured to:

obtain a set of images for training that includes a plurality of images for training and a plurality of correct data items corresponding to the plurality of images for training;

train a second neural network using the image set for training, to output, from each of the plurality of images for training inputted, the plurality of first scalars corresponding to the plurality of classes, the second neural network being a classification model;

perform a first transform that transforms a plurality of second scalars into a plurality of classification probability values, the plurality of second scalars corresponding to the plurality of classes and being obtained by inputting an image for evaluation into the second neural network trained;

determine a plurality of second classification thresholds based on the plurality of classification probability values;

perform a second transform that is a transform from the plurality of second classification thresholds into the plurality of first classification thresholds and an inverse transform of the first transform; and

output the plurality of first classification thresholds into which the plurality of second classification thresholds have been transformed by the second transform to the information processing device.

2. The information processing system according to claim 1,

wherein the first transform is a computation by an inverse-transformable probability function, and

the second transform is a computation by an inverse function of the inverse-transformable probability function.

3. The information processing system according to claim 1,

wherein the first transform is a transform using a database corresponding to a computation by an inverse-transformable probability function, and

the second transform is a transform using a database corresponding to a computation by an inverse function of the inverse-transformable probability function.

4. The information processing system according to claim 1,

wherein the image for evaluation is a set of images for evaluation including a plurality of images for evaluation,

the plurality of images for evaluation are inputted into the second neural network trained, to obtain, for each of the plurality of images for evaluation, a plurality of classification probability values corresponding to the plurality of classes, and

the plurality of second classification thresholds are determined based on a classification result using the plurality of second classification thresholds based on the plurality of classification probability values obtained for each of the plurality of images for evaluation.

5. The information processing system according to claim 1,

wherein

the plurality of classes include a class corresponding to a person, a class corresponding to a vehicle, and a class different from the class corresponding to the person and the class corresponding to the vehicle.

6. An information processing method,

wherein the information processing method causes a first processor of an information processing device to:

obtain an image captured by a camera;

obtain a plurality of first classification thresholds outputted from a threshold calculating device including a second processor, the plurality of first classification thresholds being a plurality of classification thresholds that are for classifying an object included in the image into at least one of a plurality of classes and correspond to the plurality of classes;

compare a plurality of first scalars and the plurality of first classification thresholds, wherein the plurality of first scalars are obtained by inputting the image into a first neural network that is a trained first classification model, wherein the plurality of first scalars correspond to the plurality of classes, wherein the plurality of first scalars are a plurality of scalars that are not transformed into classification probability values, and wherein the plurality of first classification thresholds correspond to the plurality of classes;

classify the object into, among the plurality of classes, at least one class having a corresponding first scalar that is greater than a corresponding first classification threshold; and

output, as a classification result, an information relating to the at least one class into which the object is classified,

wherein the information processing method further causes the second processor to:

obtain a set of images for training that includes a plurality of images for training and a plurality of correct data items corresponding to the plurality of images for training;

train a second neural network using the image set for training, to output, from each of the plurality of images for training inputted, the plurality of first scalars corresponding to the plurality of classes, the second neural network being a classification model;

perform a first transform that transforms a plurality of second scalars into a plurality of classification probability values, the plurality of second scalars corresponding to the plurality of classes and being obtained by inputting an image for evaluation into the second neural network trained;

determine a plurality of second classification thresholds based on the plurality of classification probability values;

perform a second transform that is a transform from the plurality of second classification thresholds into the plurality of first classification thresholds and an inverse transform of the first transform; and

output the plurality of first classification thresholds into which the plurality of second classification thresholds have been transformed by the second transform to the information processing device.

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